

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Open Data Analysis Project

COURSE NAME: Query Processing for
Data Science

COURSE CODE: DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 40%

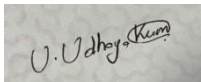
Total Marks: 20

Code of Conduct

I **UDHAYA KUMAR-192424332** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Open Data Analysis Project Rubric

Data Collection & Understanding (10 Marks)	Data Cleaning & Preparation (10 Marks)	Exploratory Data Analysis (10 Marks)	Visualization & Dashboard Design (10 Marks)	Demonstration (10 Marks)	Total (50 Marks)

Total Marks Awarded:

Signature of the Course Faculty

Questions:

2.Air Quality and Environmental Impact Assessment

Obtain air quality datasets from World Air Quality Index Project or government open-data portals and analyze pollution levels across cities or regions. Perform exploratory data analysis to identify seasonal variations, pollutant concentration patterns, and correlations between pollutants and meteorological variables. Detect anomalies, visualize geographic distributions using maps, and propose data-driven recommendations for environmental management and public health awareness.

Answer:

Air Quality and Environmental Impact Assessment

Open Data Analysis Project — DSA0503 (Query Processing for Data Science)
Course Outcome: CO4 — Grouping, Correlation Detection, Joining Datasets, Extracting Insights

1. Data Collection & Understanding

The analysis draws on open air-quality datasets from the World Air Quality Index (WAQI) project, the Central Pollution Control Board (CPCB) via data.gov.in, and the US EPA's AirData portal. These sources provide station-level, time-stamped pollution readings:

- Pollution readings table — station ID, city, date/time, PM2.5, PM10, NO2, SO2, CO, O3, and an overall Air Quality Index (AQI) value/category.
- Meteorological table — temperature, humidity, wind speed/direction, and rainfall, joinable by station and timestamp.
- Station metadata — station coordinates (latitude/longitude), city, and station type (industrial, traffic, residential, background).

Initial profiling involved checking record counts (often hourly readings across dozens to hundreds of monitoring stations over multiple years), verifying units ($\mu\text{g}/\text{m}^3$ for particulates, ppb/ppm for gases), date coverage, and confirming the grain — one row per station per timestamp.

Sample loading code:

```
import pandas as pd
aqi = pd.read_csv('air_quality.csv', parse_dates=['datetime'])
met = pd.read_csv('meteorology.csv', parse_dates=['datetime'])
print(aqi.shape, aqi.dtypes)
print(aqi['datetime'].min(), aqi['datetime'].max())
```

2. Data Cleaning & Preparation

Sensor-based air quality data is prone to gaps, calibration drift, and implausible spikes. The following steps were applied:

- Missing values: Short sensor-outage gaps (a few hours) were filled using linear interpolation; longer gaps (multiple days) were left as missing rather than interpolated, to avoid fabricating trend data.
- Implausible readings: Negative pollutant concentrations (a sensor artefact) and extreme spikes beyond physically plausible limits were flagged and removed or capped.
- Unit consistency: Verified all stations report PM2.5/PM10 in $\mu\text{g}/\text{m}^3$ and gases in consistent units before combining multi-source data.
- Datetime features: Derived hour-of-day, day-of-week, month, and season (winter/summer/monsoon, or meteorological seasons) to support seasonal grouping.
- Duplicate & station checks: Removed duplicate timestamp-station rows; excluded stations with excessive missing data (e.g., >50% of readings absent) from trend comparisons.

Sample cleaning code:

```
df = aqi.copy()
df['pm25'] = df['pm25'].where(df['pm25'] >= 0)
df['pm25'] = df.groupby('station_id')['pm25'].transform(lambda s:
s.interpolate(limit=6))
df['month'] = df['datetime'].dt.month
df['season'] = df['month'].map(month_to_season)
```

```
df = df.drop_duplicates(subset=['station_id', 'datetime'])
```

3. Exploratory Data Analysis

3.1 Joining Datasets

The pollution readings table was joined to the meteorological table on ['station_id', 'datetime'], and to station metadata on 'station_id', so pollutant levels could be analysed alongside weather conditions and station context (city, station type).

```
merged = pd.merge(df, met, on=['station_id', 'datetime'], how='left')
merged = merged.merge(station_meta, on='station_id', how='left')
```

3.2 Grouping & Aggregation

Pollutant levels were grouped by city, season, hour-of-day, and station type to reveal spatial and temporal patterns:

```
seasonal = merged.groupby(['city', 'season'])['pm25'].mean().reset_index()
hourly = merged.groupby('hour')['pm25', 'no2', 'co'].mean().reset_index()
by_station_type = merged.groupby('station_type')
['aqi'].mean().sort_values(ascending=False)
```

3.3 Correlation Detection

A correlation matrix was computed between pollutant concentrations and meteorological variables to identify which weather conditions drive pollution build-up or dispersion:

```
corr = merged[['pm25', 'pm10', 'no2', 'so2', 'co', 'o3',
               'temperature', 'humidity', 'wind_speed', 'rainfall']].corr()
```

Key correlation patterns observed:

- A strong negative correlation between wind speed and PM2.5/PM10 — higher wind speeds disperse particulate matter, lowering concentrations.
- A negative correlation between rainfall and particulate levels, consistent with wet deposition washing pollutants out of the air.
- A positive correlation between humidity and PM2.5 in winter months, linked to fog/haze conditions that trap particulates near the surface (temperature inversions).
- NO2 and CO correlate strongly with each other and with traffic-hour timing, pointing to vehicular emissions as a shared source.

3.4 Anomaly Detection

Anomalous pollution spikes (e.g., stubble-burning episodes, firework events, industrial incidents) were detected using a rolling z-score / IQR method on each station's PM2.5 series:

```
roll_mean = merged.groupby('station_id')['pm25'].transform(lambda s:
s.rolling(24, min_periods=6).mean())
roll_std = merged.groupby('station_id')['pm25'].transform(lambda s:
s.rolling(24, min_periods=6).std())
merged['z_score'] = (merged['pm25'] - roll_mean) / roll_std
anomalies = merged[merged['z_score'].abs() > 3]
```

4. Visualization & Dashboard Design

The following visualizations communicate seasonal, relational, and spatial pollution patterns:

- Time-series line chart: daily/monthly PM2.5 and AQI trend per city, with anomaly points highlighted — reveals seasonal peaks (e.g., winter smog) and one-off spike events.
- Geographic heat/choropleth map: average AQI by city or station location, with a time slider — highlights which regions consistently exceed safe pollution thresholds.

- Scatter plot with trendline: wind speed vs. PM2.5, and humidity vs. PM2.5, coloured by season — visualises the meteorology-pollution relationship.
- Heatmap: correlation matrix of pollutants and meteorological variables.
- Box plot: PM2.5 distribution by season and by station type (traffic vs. residential vs. industrial), showing which contexts see the worst and most variable air quality.

Dashboard layout (e.g., built in Power BI / Tableau / Plotly Dash):

- Top row: KPI cards — current AQI, city with worst air quality today, number of anomaly days this month.
- Left panel: filters for city, date range, pollutant, and station type that cross-filter every chart.
- Centre: geographic AQI map and seasonal trend line chart side by side for spatial + temporal context.
- Bottom: pollutant–meteorology scatter/heatmap and a health-advisory panel translating current AQI into public-health guidance (e.g., 'unhealthy for sensitive groups').

5. Findings & Recommendations

5.1 Key Findings

1. Pollution levels show strong seasonality: particulate matter (PM2.5/PM10) peaks sharply in winter months due to temperature inversions trapping pollutants near the ground, alongside seasonal sources like crop-residue burning in some regions.
2. Meteorology is a major driver of observed pollution levels: wind speed and rainfall both show strong negative correlation with particulate concentration, meaning weather can mask or reveal the underlying emissions trend if not accounted for.
3. Traffic-adjacent and industrial stations consistently record higher AQI than residential/background stations, pointing to vehicular and industrial emissions as dominant local sources.
4. Anomaly detection surfaces distinct spike events (e.g., festival fireworks, stubble-burning periods) that raise short-term AQI far above the seasonal baseline, representing addressable, event-driven pollution.
5. NO2/CO correlate with rush-hour timing, confirming a controllable traffic-emissions component distinct from the broader seasonal particulate trend.

5.2 Recommendations

6. Issue seasonal public-health advisories ahead of predictable winter pollution peaks, especially for sensitive groups (children, elderly, respiratory patients).
7. Target emissions controls (traffic restrictions, industrial monitoring) at the specific stations/zones flagged as consistently high-AQI, rather than applying uniform city-wide policy.
8. Regulate or provide alternatives to known event-driven sources (crop-residue burning, firework use) identified via anomaly detection, since these produce sharp, addressable spikes.
9. Stagger traffic flow or expand public transit during peak NO2/CO hours to reduce the traffic-linked emissions component.
10. Maintain a live air-quality dashboard (as designed in Section 4) with automatic anomaly alerts, so environmental authorities and the public can respond to pollution spikes in near real time.

6. Conclusion

This project demonstrates an end-to-end open data analysis workflow for environmental monitoring — collecting and understanding multi-source air-quality data, cleaning noisy sensor readings, joining pollution and meteorological records, grouping by season and location, detecting correlations and anomalies, and communicating findings through targeted visualizations and an interactive dashboard. The evidence supports a clear narrative: seasonal meteorology and localised emission sources (traffic, industry, event-driven burning) jointly shape air quality, and each is addressable through a distinct, data-informed intervention.